Icarus

Mission software and a validation harness for autonomous drones

Icarus gives operators a way to express a drone mission in plain language while keeping flight control, safety limits and emergency recovery outside the AI model. It is built for teams that need mission software they can inspect, test and adapt to their airframe.

What we are building

A software layer that connects an operator's mission to an existing autopilot. A local decision agent reads current vehicle state and sensor evidence, proposes a bounded action, and receives measured results before the next decision. Icarus verifies every action through a typed Drone API, guardrails, an independent safety supervisor and ArduPilot.

The software and harness

- Mission agent with compact context, typed actions and feedback after each action.
- State, health, camera, depth and LiDAR interfaces that remain separate from flight authority.
- Local LiDAR-based route detours, geofence checks, guardrails and safe hold, return or land recovery.
- Gazebo and AirSim simulation tracks, ArduPilot SITL, fault injection, sealed episode logs and replay.

Where it can be useful

Defence and security

- Operator-supervised perimeter observation and reconnaissance.
- Mission rehearsal in the same software harness used for flight testing.
- Replayable evidence for mission review, training and failure analysis.

Enterprise operations

- Industrial inspection, site security and asset surveys.
- A reusable mission layer across drone OEMs, sensors and autopilot setups.
- Repeatable safety, fault and scenario testing before field deployment.

Startup direction

Icarus can become the integration and validation layer for drone manufacturers, systems integrators and operators. The initial offer is mission software, simulator-based qualification and customer-specific workflow integration, followed by controlled pilot deployments.

Current stage: simulation-first R&D. The AI mission layer is not approved for unattended or physical flight.